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Compositions and methods for gene targeting using CRISPR-Cas and transposons

Abstract

Provided are compositions and methods for transposon-CRISPR-based DNA editing. The compositions and methods include a modified TnsA protein that is altered from its original sequence found in *Aeromonas salmonicida*, and provides improved transposition frequency, and is functional in a heterologous system. The TnsA protein is used in a system with transposon proteins TnsB, TnsC, TniQ, Cas proteins Cas8f, CasSf, Cas7f, Cas6f, a transposable DNA cargo sequence that is flanked by left and right transposon sequences, and at least one guide RNA that contains at least one spacer targeted to a target DNA sequence in the chromosome or the extrachromosomal element.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application claims priority to U.S. provisional patent application No. 62/818,523, filed Mar. 14, 2019, and U.S. provisional patent application No. 62/882,204, filed Aug. 2, 2019, the entire disclosures of each of which are incorporated herein by reference.

FIELD

(1) The present disclosure relates generally to approaches for targeting genes and more particularly for improved compositions and methods using combinations of Tn7-like transposons with Cas8/5f, Cas7f, and Cas6f for improved and flexible chromosome editing.

BACKGROUND

(2) Existing CRISPR-Cas systems can similarly be used to deliver DNA to a specific site in a single orientation. However, these processes involve a cumbersome two-step process where the CRISPR-Cas system and guide RNA are first used to break the host chromosome. In a second process, a DNA substrate that must be introduced at the same time must then be used by another form of DNA recombination to repair the DNA break by replacing it with an investigator introduced DNA substrate. This recombination process is often a natural host recombination system but sometimes another synthetic system expressed in the host. This is an inefficient process that can also lead to unwanted off-target events. There is accordingly an ongoing and unmet need for improved CRISPR-based approaches to chromosome editing. The present disclosure is pertinent to this need.

SUMMARY OF THE DISCLOSURE

(3) The present disclosure provides compositions and methods for modifying chromosomes and extra-chromosomal elements, the latter including but not necessarily limited to plasmids, viral or other pathogen-derived DNA, and DNA present in, for example, organelles in eukaryotic cells, such as mitochondria and chloroplasts. DNA modified according to the present disclosure may be cytoplasmic or nuclear DNA, and acts on double stranded DNA. The disclosure includes modification of double-stranded DNA templates without inducing double-stranded breaks, and thus in embodiments comprises transposing a segment of a double stranded DNA to another location, in cis or in trans.

(4) The disclosure is illustrated in non-limiting embodiments by demonstrating transposition from a prokaryotic chromosome to a plasmid, and transposition from one location in a prokaryotic chromosome to another location in the same chromosome. It is accordingly considered that the system is readily adaptable for use in eukaryotic cells and organisms, using the same set of proteins and/or vectors encoding the proteins, by configuring the proteins as needed to be located in the nucleus of eukaryotic cells.

(5) The disclosure, at least in part, provides for increased transposition efficiency, relative to suitable controls, as described further below. In embodiments, the disclosure includes use of a modified TnsA protein for which an unmodified counterpart was initially identified as encoded by *Aeromonas salmonicida* strain S44, on its plasmid pS44-1. By analysis that is described more fully below, and without intending to be bound by any particular theory, it is considered that the wild type TnsA protein encoded by the pS44-1 plasmid is non-functional in its native environment. The disclosure accordingly includes a modified version of this protein that is functional in the presently described system, and moreover is functional in at least one heterologous system, illustrated in non-limiting embodiments using *E. coli*. Thus, the disclosure provides, in one embodiment, a method for modifying a chromosome or extrachromosomal element in one or more heterologous cells.

(6) A method of the disclosure generally comprises introducing into one or more cells a TnsA transposon protein that is at least 90% similar to SEQ ID NO:1 but contains an amino acid other than Alanine at position 125 of SEQ ID NO:1, or by introducing into the cells an expression vector comprising a tnsA gene that encodes and expresses the modified TnsA transposon protein when introduced into one or more cells. The modified TnsA protein functions in the context of transposon proteins TnsB, TnsC, TniQ, and Cas proteins Cas8f, Cas5f, Cas7f, Cas6f, examples of which are known in the art and are adaptable for use with the presently provided system. The method also comprises a transposable DNA cargo sequence that is flanked by left and right transposon sequences, said left and right transposon sequences being readily recognizable by those skilled in the art, and at least one guide RNA comprising a spacer targeted to a target DNA sequence in the chromosome or the extrachromosomal element. The DNA target sequence comprises a protospacer and a protospacer adjacent motif (PAM) that is 5' to the protospacer. Thus, the spacer in the guide RNA may be complementary to a protospacer. By providing cells that comprise each of these components, the disclosure facilitates the guide-RNA directed transposition of any desired DNA cargo into the chromosome or extrachromosomal element in a location selected by the user of this system. The location of the transposition is generally 3' relative to the PAM and the protospacer sequence, following established principles of Tn7 transposition events with respect to nucleotide positions that are known in the art.

(7) In certain implementations, the TnsA transposon protein is at least 90% similar to SEQ ID NO:1 and comprises an amino acid that is not Alanine at position 125 of SEQ ID NO:1. In a non-limiting embodiment, the TnsA transposon protein comprises a D at position 125 of SEQ ID NO:1, or a sequence that is at least 90% identical to SEQ ID NO:1, with the proviso that position 125 is not Alanine. In embodiments, the TnsA protein comprises SEQ ID NO:2, which includes a A125D change, relative to the amino acid sequence of SEQ ID NO:1.

(8) In embodiments, a chromosome or extrachromosomal element that is modified according to the present disclosure is present in one or more prokaryotic cells. In embodiments, a chromosome or extrachromosomal element that is modified according to the present disclosure is present in one or more eukaryotic cells. In embodiments, efficiency of transposition of the transposable DNA cargo in a population of cells is more efficient than transposition obtained from a control value. In embodiments, the control value is obtained using a TnsA transposon protein that comprises an amino acid sequence that is at least 90% similar to SEQ ID NO:1 and contains an Alanine at position 125.

(9) The disclosure includes introducing the described proteins, DNA encoding the proteins, and ribonucleoprotein complexes, directly into cells. In embodiments, the disclosure thus includes introducing into cells one or more expression vectors that encode at least one of the TnsA, TnsB, TnsC, TniQ, Cas8f, Cas5f, Cas7f, and Cas6f proteins. The disclosure also includes modifying DNA in cells that endogenously express one or more of these proteins, provided at least one of the proteins is introduced into the cells. Likewise, the DNA cargo to be transposed in a guide-RNA directed manner may be provided as one or more distinct molecules, or may be present on one or more of the expression vectors, or by a distinct vector, or may be pre-existing in the cell or cells to be modified, i.e., the disclosure includes transposition of an endogenous DNA element. In non-limiting embodiments, the transposable DNA cargo sequence is transposed into the chromosome or extrachromosomal element within 42-52 nucleotides 3' relative to the 3' end of the protospacer. In embodiments, the transposable DNA cargo is inserted in an orientation with the right end proximal to the protospacer. In embodiments, this orientation occurs in all insertions in a population of cells.

(10) Expression vectors encoding one or more proteins described herein are included in the disclosure, as are cells that comprise such expression vectors. The disclosure includes such expression vectors in combination with a transposable DNA cargo sequence that is flanked by left and right transposon sequences. Cells modified using the compositions and methods of the disclosure are also included.

(11) Also provided are kits for use in a method of this disclosure. The kit includes an expression vector encoding a TnsA transposon protein that is at least 90% similar to SEQ ID NO:1 but contains an amino acid other than Alanine at position 125 of SEQ ID NO:1. The kit may also include one or more expression vectors that encode one or a combination of TnsB, TnsC, TniQ, Cas8f, Cas5f, Cas7f, and cas6f. The kit may also include an expression vector configured for accepting a sequence encoding a suitable guide RNA to facilitate RNA-guided transposition of any desired DNA cargo to any desired location.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1. Cartoon depiction illustrating Tn7-like CRISPR-Cas system targeting transposon insertions adjacent to matches to the guide RNA. The Guide RNA is encoded between repeats in the CRISPR array on the element. Insertions occur in one orientation based on the Left (L) and Right (R) end-sequences of the transposon.

(2) FIG. 2. Illustration of a representative embodiment of the disclosure. In the illustrated system, the transposon (Tn) and Cas proteins as described herein are expressed, for example, from a plasmid. These proteins catalyze the movement of the genes of interest, such as DNA cargo, into a selected target in the bacterial genome, or on a plasmid. FIG. 2—Panel A—Transposon and Cas proteins are expressed in bacteria or other cell type where the DNA insertion is desired. A guide RNA is designed such that it matches the position of interest in the cell. FIG. 2 Panel B. Illustration of expression of the guide RNA and the DNA to be delivered are shown in a non-limiting embodiment on a separate plasmid. DNA oligonucleotides encoding the guide RNA are shown as

cloned into a synthetic CRISPR array on the plasmid. The desired genes or other DNA cargo to be delivered into this site are cloned into the vector using standard molecular techniques using a multicloning (MCS) site located between the left and right end transposon sequences in the vector. The delivery vector facilitates expression of the guide RNA in the target host cell(s). In one variation of this procedure, such as in prokaryotes, the plasmid will not be maintained, but will allow the genes of interest or other DNA to be delivered to the target site recognized by the guide RNA. For ease of selecting the desired insertion, a drug resistance marker can also be included in the vector (DrugR).

(3) FIG. 3. Representative photograph of electrophoretic separation and sizes of Cascade proteins produced recombinantly. In non-limiting embodiments, proteins derived from, for example, *Aeromonas* strains, can be produced in *E. coli* using different promoters from separate vectors, in individual cells or jointly in the same strain background. Markers designate kDa.

(4) FIG. 4. Representative photograph of Immobilized Metal Ion Affinity Chromatography (IMAC) copurification of cascade complex. This figure demonstrates that by selectively isolating one of the components of the Cas complex and co-expressing the guide RNA, a complex with all three subunits is formed and can be pulled down using IMAC.

(5) FIG. 5. Representative photograph of SDS-PAGE with Coomassie staining demonstrating that the Cas8-5, Cas7, Cas6, and guide RNA complex can be pulled down with an affinity tag either on Cas8-5 or on Cas6 and can be retained in a spin purification column with a molecular weight cut-off of 100 kiloDaltons. Complexes are shown above the image as cartoon representations.

(6) FIG. 6. Representative photograph of electrophoretic separation of DNA and protein complex with guide RNA and showing that a specific complex is found with target DNA with PAM+protospacer that is not found with a no protospacer control. To generate the results shown in the figure, a crude preparation Cas8-5, Cas7, Cas6-HIS6 $\times$, and guide RNA was incubated with a 200 bp double-stranded DNA target DNA (with or without the PAM+protospacer). A product that moves faster than the guide RNA complex with the PAM+protospacer indicates that an R-loop has formed, an important transition indicated the protospacer was fully recognized. This is not found with the “no protospacer” control. 100 ng of supercoiled plasmid DNA was added as a competitor for non-specific binding and an oligonucleotide specific to the displaced strand to help stabilize the complex. Products were separated in an agarose gel as indicated and stained with Ethidium Bromide for visualization.

(7) FIG. 7. Representative photograph of electrophoretic separation using SDS-PAGE with Coomassie staining. The results demonstrate that Cascade assembles and is stable as a complex when produced in *E. coli* in vitro. This demonstrates the CRISPR genes can process precrRNA and form cascade complex in heterologous host. Complexes are illustrated above the image as cartoons.

(8) FIG. 8. Representative photograph of a mobility shift assay where the DNA substrate was visualized using a Cy3 fluorophore with excitation at the appropriate wavelength. The TniQ, Cas8-5, Cas7, Cas6 with guide RNA complex was incubated with a Cy3 labeled DNA substrate and then loaded onto a polyacrylamide gel to determine if the substrate was recognized and specifically bound. Only a substrate that included the PAM and protospacer was specifically recognized. The result shows that Cascade binds protospacer DNA specifically.

(9) FIG. 9. Cartoon depiction of functional assay showing the contribution of TniQ.

(10) FIG. 10. Representative photograph of electrophoretic separation of complexes. The result shows that Cascade is stable as a complex as produced in *E. coli* and interacts with TniQ and can be separated out of solution using a HIS6 affinity tag. Dotted line indicates the altered position with and without the His6 affinity tag.

(11) FIG. 11. Representative photograph of mobility shift assay as a functional assay showing that the behavior of the complex (Cas8-5,7,6) has more/better binding ability with TniQ. The results also show the complex specifically binds a protospacer target DNA (Pt), but not a randomly selected other DNA target (nPt). The gel represents a mobility shift assay on 3.5% sodium borate

polyacrylamide gel, pH7.5. DNA substrates were visualized using a Cy3 fluorophore with excitation at the appropriate wavelength.

(12) FIG. 12. Representative photograph of assay demonstrating specific and non-specific binding based on distinct PAM sequences. These results show that even the same guide RNA is only strongly recognized when the protospacer/spacer-match has the TACC (wt) PAM, but not the changes shown. The results also show that 4 mismatches/changes in the “seed” region (the region closer to the PAM) appear to not allow recognition, but 4 changes in the distal region more distant from the PAM appear to be tolerated. These results demonstrate strong specificity with the functioning with the guide RNA complex.

(13) FIG. 13. Cartoon scheme of the transposition assay, and specifically depicting targeting of an F plasmid derivative with a candidate protospacer target. Expression plasmids contain TnsABC, TniQ/Cascade, and a synthetic CRISPR array. The donor element is a single copy inserted into chromosome. The figure shows the design of an experiment performed in a heterologous host (*E. coli*) as demonstrated in the Examples. The *plac* and *pBAD* are inducible promoters. The mini element is initially present in the *E. coli* chromosome, and guide RNA-directed transposition directed to the specific place next to the protospacer/spacer-match, which is measured by mating the conjugal/mobile F plasmid into another *E. coli* strain and the frequency determined (shown in FIG. 14).

(14) FIG. 14. Carton depiction, expanding on FIG. 13. The image depicts Guide RNA-directed transposition reconstituted in *E. coli* using an adaptation of a standard mate out assay. The image depicts how frequency of conjugal plasmids that now contain an insertion event are monitored by mating a population of cells post-induction with a Nal.^{sup}.R Rif.^{sup}.R recipient strain, followed by selection for recipient cells with the conjugal plasmid and mini element markers. In representative example, the full suite of transposition proteins resulted in thousands of colonies while the control lacking TnsABC had none. Frequency is estimated at least 10.^{sup}.-2 in these conditions and has been determined to be higher, as described in the Examples.

(15) FIG. 15. Cartoon depiction of representative guide RNA-directed transposition reconstituted in *E. coli* with photographic representation of results. PCR across the predicted target site confirms transposition is not occurring elsewhere in the conjugal plasmid, but specifically next to the protospacer with 4 examples which were also sequenced. PCR with left and right end primers with 8 isolates were all in the correct orientation as predicted (not shown). DNA sequencing confirmed that all four insertions that were analyzed were correct with target site duplication found with transposition. Insertions were at 47, 48, 48, 49 bp from the protospacer. In this example, only protospacers+10 bp from *Aeromonas* is on each side in *E. coli*, demonstrating the insertions were made in new, engineered sequences in the heterologous host. The protospacer containing target is indicated with crosshatching. Antibiotic resistance makers for Kanamycin resistance (KanR) and Spectinomycin resistance (SpecR) used to identify the F plasmids with transposition events are indicted.

(16) FIG. 16. Graphical data obtained with the mate-out assay depicted in FIGS. 13 and 14. The results demonstrate that transposon insertions can be programed to insert into either a *lacZ* targeted (depicted in the cartoon) or a portion of the native *ffs* gene native target found in *A. salmonicida* (not cartooned). All of the protospacer targets have the TACC PAM. Targets were selected for the *lacZ* gene as specific for the top or bottom strand and the relative position indicated (it is unclear why an ~10-fold difference in frequency is found with different protospacers). The native *ffs* guide RNA (*ffswild*-type) has 10 mismatches to the native *ffs* gene. Making the guide RNA a perfect match (*ffsexact*) increases the frequency of transposition.

(17) FIG. 17. Graphical data demonstrating that TnAs allows RNA-guided transposition to chromosomal *lacZ*. Expression of TnsABC (TnsA A125D), TniQ, Cas8/5f, Cas6f, Cas7f, and single repeat-spacer-repeat array units in *E. coli* allows RNA-guided transposition to *lacZ* as monitored by Lac phenotype on MacConkey agar. Transposition can be programmed with synthetic

spacers to lacZ with two randomly selected TACC PAM protospacer sequences on either strand (the same lacZ specific guider RNAs used in FIG. 19). The actual data used to calculate the percentages shown in the graph from the three replicates is indicate below the graph.

(18) FIG. 18. Depiction of relationship between spacer and protospacer with the gene targeted and the position of the insertion. Five different genes targeted by guide RNAs are indicated by the spacers found in the array. Matches between the guide RNA and protospacer are indicated by a short vertical line. The five base-pair target site duplication (TSD) is indicated along with the distance from the protospacer recognized by the guide RNA to. The right end of the transposon is indicated. The RNA lacZ3 is SEQ ID NO:15, The RNA lacZ4 is SEQ ID NO:17. The RNA RNA ffs.sup.exact is SEQ ID NO: 12. The uppermost DNA strand is SEQ ID NO:19. The middle DNA strand is SEQ ID NO:20. The bottom DNA strand is SEQ ID NO:21.

DETAILED DESCRIPTION

(19) Unless defined otherwise herein, all technical and scientific terms used in this disclosure have the same meaning as commonly understood by one of ordinary skill in the art to which this disclosure pertains.

(20) Every numerical range given throughout this specification includes its upper and lower values, as well as every narrower numerical range that falls within it, as if such narrower numerical ranges were all expressly written herein.

(21) The disclosure includes all polynucleotide and amino acid sequences described herein. Each RNA sequence includes its DNA equivalent, and each DNA sequence includes its RNA equivalent. Complementary and anti-parallel polynucleotide sequences are included.

(22) Every DNA and RNA sequence encoding polypeptides disclosed herein is encompassed by this disclosure. Amino acids of all protein sequences and all polynucleotide sequences encoding them are also included, including but not limited to sequences included by way of sequence alignments. Sequences of from 80.00%-99.99% identical to any sequence (amino acids and nucleotide sequences) of this disclosure are included.

(23) The disclosure includes all polynucleotide and all amino acid sequences that are identified herein by way of a database entry. Such sequences are incorporated herein as they exist in the database on the filing date of this application or patent.

(24) The disclosure includes polynucleotides encoding proteins of this disclosure, and polynucleotides that comprise transposons, including but not necessarily limited to Tn7-like transposons. In certain approaches of this disclosure expression vectors, such as plasmids, are used to produce one or more than one construct and/or component of the system, and any of their cloning steps or intermediates. A variety of suitable expression vectors known in the art can be adapted to produce components of this disclosure.

(25) In embodiments, the disclosure provides adaptations of type I-F CRISPR-Cas systems. In embodiments, the disclosure provides adaptations of Type I-B CRISPR-Cas systems. In embodiments, the disclosure provides adaptations of Tn7-like transposons. In particular, bacterial genomes shows that many Tn7-like transposons contain ‘minimal’ type I-F CRISPR-Cas systems that contain of fused cas8f and cas5f, cas7f and cas6f genes, and a short CRISPR array. Additionally, several small groups of Tn7-like transposons encompass similarly truncated type I-B CRISPR-Cas systems. This gene composition of the transposon-associated CRISPR-Cas systems implies that they are competent for pre-crRNA processing yielding mature crRNAs and target binding but not target cleavage that is required for interference. Accordingly, in developing aspects of the present disclosure, e.g., to identify elements that could potentially be adapted for use in the presently described approaches, approximately 400 elements were initially analyzed by searching for elements that are highly represented with uninterrupted genes and contiguous right and left transposon end sequences where closely related transposons were found in separate species. These elements from hosts in the family Vibrionaceae, were assembled in expression systems that are regulatable in *E. coli*, and were tested for guide RNA-directed transposition activity. This approach

did not provide any reliable evidence that transposition activity had occurred. A significant amount of additional analysis and experiments were carried out by altering expression construct promoters, translation signals, and general operon structure, and to establish genetic systems where low levels of transposition could be detected, but where background was also low to minimize false positives. This approach also did not identify evidence of guide RNA-directed transposition activity using these systems in heterologous hosts. The possibility that host factors endogenous to the original host were required was then analyzed. In particular, experiments were carried out to integrate promoters into the native host to drive expression of the operons to determine evidence of transposition. These experiments also gave no evidence of detectable transposition. It was accordingly unclear if the analyzed elements were inactivated by, for example, unknown missense mutations, or only active during a special cycle of growth, or in the presence of a special inducing factor, or naturally transposed at a frequency lower than the detection limit of the system. In response to these failed attempts to identify components that could be used in the presently described system, a more advanced method was employed. Specifically, attempts were made to identify candidate Tn7-CRISPR/Cas elements with the type I-F variant systems from a larger pool of sequences. This approach was taken in part to identify potential candidate elements that could be deliberately modified so that they would be suitable for genome editing by guide RNA-directed transposition, particularly in heterologous hosts. To this end, we analyzed approximately 300,000 genomes, and in turn focused on the gammaproteobacterial class of bacteria. This latter approach included analysis of approximately 45,000 genomes from distinct bacterial types, one non-limiting example of which included all transposon proteins from *Aeromonas hydrophila*, including but not limited to its TnsA protein. Specifically, we analyzed the sequence of gammaproteobacterial genomes for the signature proteins of TnsA, TnsB, TnsC, TniQ/TnsD, Cas8/5f, Cas7f, and Cas6f using the definitions in the pfam database (pfam.xfam.org/). The gene location information was used to computationally determine the convergence of these genes with the expected operon structure. This led to a collection of just under 800 candidate Tn7-CRISPR/Cas elements with the I-F variant systems. The majority of elements were found to reside in species from within the bacterial family Vibrionaceae within the gammaproteobacteria, which, without intending to be bound by any particular theory is to be considered to be where they likely originated. Based in part on this analysis, we focused on elements found outside the Vibrionaceae in view of our hypothesis that they may have accumulated alleles that made them less dependent on host factors. We therefore attempted to identify and refine a system that was not found in a Vibrionaceae.

(26) A computationally-assisted decision matrix was developed that took into account criteria including location where the bacterial host was originally isolated, phylogenetic relatedness of the bacterial hosts, and amino acid variability weighed by the domain structure and active sites within the seven Tns/Tni/Cas proteins. We also took into account that alleles with increased activity often come with host fitness costs, selecting for secondary and confounding moderating alleles. This analysis identified the genus *Aeromonas* as a potential source of elements that could be modified for use in the presently described system. We produced a number of candidate alleles which could be tested for the capability to cause guide RNA-directed transposition in a heterologous system, using as a non-limiting demonstration the non-native *E. coli* host. Based at least in part on this analysis, the pool of alleles was used to identify a number of potential amino acid changes in *Aeromonas* proteins that could potentially convert a system that does not exhibit desired guide RNA-directed transposition functionality in its native host, let alone in a heterologous host, into a system that exhibits desired guide RNA-directed transposition functionality in a heterologous host. Accordingly, the present disclosure provides, in one embodiment, a modification of the TnsA protein that is encoded by *Aeromonas salmonicida* strain S44 plasmid pS44-1. In this regard, the present disclosure shows using representative assays that a single amino acid change in this TnsA protein is necessary and sufficient to convert the presently described system from a non-functional to a functional guide RNA-directed transposition system in a heterologous host. Specifically, a

change in the TnsA protein of residue 125 from Alanine (A) to Aspartic Acid (D) was able to impart guide RNA-directed transposition functionality in the representative heterologous system of *E. coli*. This novel functionality was not expected based the genomic context of TnsA in its closest relative (*Aeromonas hydrophila*) which has the same or similar sequences and is unambiguously non-functional, in the native host. The system thus, in part, may exploit of Tn7-like transposons that naturally control target site selection, but comprise certain DNA and amino acid changes that are further described herein.

(27) In embodiments, the compositions and methods of this disclosure are functional in a heterologous system. “Heterologous” as used herein means a system, e.g., a cell type, in which one or more of the components of the system are not produced without modification of the cells/system. A non-limiting embodiment of a heterologous system is any bacteria that is not *Aeromonas salmonicida*, including but not necessarily limited to *Aeromonas salmonicida* strain S44. In embodiments, a representative and non-limiting heterologous system is any type of *E. coli*. A heterologous system also includes any eukaryotic cell.

(28) In embodiments, any protein of this disclosure may be an *Aeromonas salmonicida* strain S44 protein, or a derivative thereof, with the exception that the TnsA protein is not produced by *Aeromonas salmonicida* strain S44, without modification, such as by recombinant engineering of the type described further herein.

(29) In embodiments, the presently described systems are used to direct blocks of genes to virtually any position in a bacterial genome, any episomal element, or a eukaryotic chromosome, in an orientation dependent fashion. In embodiments, the system is thus targeted to a sequence in a chromosome in a eukaryotic cell, or to a DNA extrachromosomal element in a eukaryotic cell, such as a DNA viral genome. Thus, the disclosure includes modifying eukaryotic chromosomes, and eukaryotic extrachromosomal elements. Accordingly, the type of extrachromosomal elements that can be modified according to the presently described compositions and methods are not particularly limited.

(30) As known in the art, transposons are genetic elements that can move within a genome that appear to be found in all forms of life. As discussed above, the present disclosure relates in part to a version of the Tn7-like element where it has adapted the CRISPR-Cas system as a mechanism of targeting where the transposon moves, and further comprises mutations in certain Tn-related proteins that enhance CRISPR-Cas based editing using transposon proteins.

(31) CRISPR-Cas systems are typically naturally found as acquired immune systems in bacteria. They utilize short genetic features known as “guide RNAs” as a mechanism to destroy DNAs that invade the cell. The guide RNAs are encoded in what is known as a CRISPR array that is processed with CRISPR associated proteins (or Cas proteins) to make a complex (Cas proteins+guide RNA) to target DNAs that match the guide RNA sequence for destruction by cleaving the DNA. The naturally occurring element has evolved to use a subset of the Cas proteins (Cas8/5f, Cas7f, and Cas6f) to process a cognate CRISPR array containing the guide RNA to target a cognate Tn7-like element to direct transposition adjacent to the DNA match to the guide RNA sequence. As is known in the art, Cas8/5f (also referred to as Cas8-5) are naturally fused in I-F variant systems associated with Tn7-like elements. The transposon proteins involved in this process are TnsA, TnsB, TnsC, and TnsD/TniQ. They recognize cognate “left” and “right” transposon DNA sequences that flank the transposon. Thus, as is also known in the art, each left and right end sequence pair is ordinarily associated with a particular set of tnsA, tnsB and tnsC genes, and the left and right end sequences are considered “cognate” with respect to the particular tnsA, tnsB and tnsC cassette. In addition to the I-F systems that are modified as described herein, the disclosure includes adapting other systems, including but not necessarily limited to type I-B CRISPR/Cas systems.

(32) The present disclosure demonstrates that transposon and CRISPR-Cas systems can be used in cells to target insertion of the element into a single position adjacent to the match to the guide RNA in one orientation (see, for example, FIG. 1). This system has been recapitulated using recombinant

approaches such that the transposon proteins and Cas proteins can be expressed in any position in the cell and they will act on the CRISPR array and transposon end-sequences found elsewhere in the cell.

(33) A non-limiting and general embodiment of the disclosure is depicted in FIG. 2. Each set of genes described herein can also include a suitable *xre* gene that encodes a transcription regulator. Further, any of the *tns* genes, as further described herein, may comprise mutations such that *ns* genes encode proteins that are distinct from the proteins that are produced in nature, i.e., proteins that are produced by bacteria that have not been engineered to produce a modified Tns protein.

(34) In particular, any cell of interest can be adapted to express the transposon and Cas proteins. For bacteria, this can be from an independently replicating plasmid or bacteriophage DNA or other element, or a vector that integrates into the genome, or an alternative delivery vector that is maintained or not maintained afterwards. In one embodiment, the user designs a guide RNA that matches the sequences adjacent to the desired point of insertion. Suitable approaches for designing guide RNAs are established and take into account any sequence requirements that are dictated by any adjacent motifs (called PAM sequences). This guide RNA is cloned into a delivery vector between the CRISPR array repeats (see, for example, FIG. 2, panel B). Any guide RNA used in embodiments of this disclosure can comprise RNA sequences which comprise RNA equivalents of all or portions of repeat sequences. In embodiments, a guide RNA of this disclosure may be processed by one or more proteins described herein. In embodiments, the guide RNA is processed from a longer RNA polynucleotide that comprises, for example, at least one repeat-spacer-repeat segment. More than one such segment may be included in the RNA that is processed into the guide RNA, and each processed guide RNA may be configured to target the same DNA sequence, or distinct DNA sequences.

(35) The disclosure includes using a least one *tniQ* gene, and accordingly two or more different *tniQ* genes may be used. *tniQ* genes produce a TniQ protein that is an optional part of the present system. Including this gene in the construct will direct transposition event into the one specific cognate site recognized by the TniQ protein. Without intending to be bound by any particular theory, it is considered that TniQ may also interact with the CRISPR/Cas and be required for guide RNA targeting. The genes of interest that are to be delivered into the bacterial strain are cloned into a multicloning site (MCS) in the delivery vector using existing standard lab techniques (FIG. 2, panel B). The MCS is located between the left (L) and right (R) synthetic transposon end-sequences. If orientation of the final insertion is important for a particular embodiment, the right end of the element will be proximal to the match to the selected guide RNA. The delivery vector can be designed as a conditional vector that will not be maintained if desired. If desired, a selectable genetic marker can also be included in this vector. If the delivery vector will not be maintained, integration of the DNA by the targeted transposition process can be directly selected. If the efficiency is high enough, then this selectable marker is not needed.

(36) This system can also be used to inactivate any gene in a prokaryotic or eukaryotic genome. Any one of many selectable markers can be included in the delivery vector to allow inactivation of a gene targeted by the guide RNA. This type of technology is broadly applicable to engineering new bacterial strains and eukaryotic cells for industry, research and therapeutic applications.

(37) In contrast to existing CRISPR-based editing techniques, one advantage of the present systems is that no separate DNA break is used; instead the DNA fragment of interest is directly joined into the genome at one position determined by the user.

(38) In non-limiting demonstrations, the disclosure shows that systems described herein, and which include recombinantly produced proteins (the Cas proteins with or without TniQ are referred to in certain instances as ‘cascade’) can specifically recognize and bind to a DNA substrate that comprises a protospacer. As used in certain examples, cascade comprises Cas8-5 (encoding fused Cas proteins), Cas7, Cas6 and a guide RNA with or without one or more TniQ proteins. This combination illustrates cascade for variant I-F systems associated with Tn7-like elements.

(39) For example, as shown in the examples and figures of this disclosure, in vitro binding of cascade occurs with specificity to a DNA substrate comprising a protospacer, to which the cascade complex is directed using a suitable guide RNA. Likewise, the figures and examples demonstrate copurification of a complex comprising TniQ and cascade. Thus, the disclosure shows that recombinantly produced TniQ and cascade form a physical association. Moreover, the disclosure demonstrates functionality of the system in a living heterologous system (illustrated using *E. coli*). In particular, at least FIG. 16 and Example 2, show guided transposition that is specific for a particular location in a conjugal plasmid, and that this transposition is PAM specific. In particular, in endogenous *Aeromonas*, the insertion was 48 base pairs from the protospacer. In the heterologous *E. coli* system, four insertions analyzed were 47, 48, 48, and 49 base pairs away from the protospacer, demonstrating an unexpected degree of precision in a heterologous system. Thus, the disclosure demonstrates functionality of the system using recombinant approaches in living cells that do not, without modification as described herein, produce a directed transposition event. Additionally, the disclosure demonstrates transposition from one location in a chromosome to another location in the chromosome, as shown in FIG. 17, and described in Example 3, results which are also obtained in a heterologous system, using *E. coli* as a representative example.

(40) In embodiments, systems of this disclosure include a DNA cargo for insertion into a eukaryotic chromosome or extrachromosomal element, or in the case of prokaryotes, a chromosome or a plasmid. Thus, instead of transposing an existing segment of a genome in the manner in which transposons ordinarily function, the disclosure provides for insertion of DNA cargo that can be selected by the user of the system. The DNA cargo may be provided, for example, as a circular or linear DNA molecule. The DNA cargo can be introduced into the cell prior to, concurrently, or after introducing a system of the disclosure into a cell. The sequence of the DNA cargo is not particularly limited, other than a requirement for suitable right and left ends that are recognized by proteins of the system. The right and left end sequences that are required for recognition are typically from about 90-150-bp in length. As is known in the art, such 90-150 bp length comprises multiple 22 bp binding sites for the TnsB transposase in the element in each of the ends that can be overlapping or spaced. The minimum length of the DNA cargo is typically about 700 bp, but it is expected that from 700 bp to 120 kb can be used and inserted. The disclosure provides for insertion of a DNA cargo without making a double-stranded break, and without disrupting the existing sequence, except for several residual nucleotides at the insertion site, as is known in the art for transposons. In embodiments, the insertion of the DNA cargo occurs at a position that is from approximately 47, 48, or 49 nucleotides from a protospacer in the target (e.g., chromosome or plasmid) sequence.

(41) In embodiments, the transposable DNA cargo sequence is transposed into the chromosome or extrachromosomal element within a 5 nucleotide sequence that includes the nucleotide that is located 47 nucleotides 3' relative to the 3' end of the protospacer. In embodiments, a DNA cargo insertion comprises an insertion at the center of a 5 bp target site duplication (TSD). Thus, by providing a suitable guide RNA that comprises a PAM that is cognate to the protospacer, precise integration of a DNA cargo can be achieved. In embodiments, the PAM comprises or consists of TACC or CC.

(42) The transposon and Cas genes can be expressed from any of a wide variety of existing mechanism that can replicate separately in the cell or be integrated into the host cell genome. Alternatively, they could be expressed transiently from an expression system that will not be maintained. It is expected that the proteins themselves could be directly transformed into the host strain to allow their function. The disclosure allows for multiple copies of distinct transposon gene cassettes, multiple copies of Cas genes, CRISPR arrays, and multiple distinct cargo coding sequences to be introduced and to modify genetic material in the same cell. In embodiments a first set of transposon genes *tnsA*, *tnsB*, *tnsC*, and optionally one or more *tniQ* genes, Cas genes *cas8f*, *cas5f* *cas7f*, and *cas6f* and an *xre* gene, and a sequence encoding at least a first guide RNA that is

functional with proteins encoded by the Cas genes, wherein at least one of the first set of transposon genes, the Cas genes, or the sequence encoding the first guide RNA are present within and/or are encoded by a recombinant polynucleotide that is introduced into bacteria, or eukaryotic cells. The disclosure thus includes second, third, fourth, fifth, or more copies of distinct transposon genes, Cas genes, and distinct cargo coding sequences.

(43) The delivery vector can be based on any number of plasmid, bacteriophage or another genetic element, when used in prokaryotes. The vector can be engineered so it is maintained, or not maintained (using any number of existing plasmid, bacteriophage or other genetic elements). Delivery of these DNA constructions in bacteria can be by conjugation, bacteriophage or any transformation processes that functions in the bacterial host of interest. Modifications of this system may include adapting the expression system to allow expression in eukaryotic or archaeal hosts. In embodiments, for eukaryotic cells, the disclosure includes use of at least one nuclear localization signal (NLS) in one or more proteins. In general, a suitable NLS includes one or more short sequences of positively charged lysines or arginines exposed on the protein surface. In embodiments, a system of this disclosure is introduced into eukaryotic cells using, for example, one or more expression vectors, or by direct introduction of ribonucleoproteins (RNPs). In embodiments, expression vectors comprise viral vectors. In an embodiment, adenoviral vectors may be used, and many such vectors are known in the art.

(44) Further modification of this approach can include expression and isolation of the proteins required for this process and carrying out some or all of the process in vitro to allow the assembly of novel DNA substrates. These DNA substrates can subsequently be delivered into living host cells or used directly for other procedures. Thus, the disclosure includes compositions, methods, vectors, and kits for use in the present approach to CRISPR—Transposon based editing.

(45) In one example, the disclosure provides a system for modifying a genetic target in bacteria and/or eukaryotic cells. The system comprises a first set of transposon genes *tnsA*, *tnsB*, *tnsC*, and optionally one or more *tniQ*, Cas genes *cas8f* *cas5f* *cas7f* and *cas6f*, and an *xre* gene encoding a transcription regulator, and a sequence encoding a first guide RNA that is functional with proteins encoded by the Cas genes, wherein at least one of the first set of transposon genes, the Cas genes, and/or or the sequence encoding the first guide RNA are present within and/or are encoded by a recombinant polynucleotide. Without intending to be constrained by any particular theory, it is considered that the *xre* gene, while annotated as a transcriptional regulator, can also make transposition complexes described herein more efficient.

(46) In embodiments, one or more of the *tns* genes, and therefore the proteins they encode, are modified, as described in more detail below. From this disclosure, and other information known to those skilled in the art, homologous proteins can be recognized, aligned, and amino acid changes in the proteins can be made such that the proteins function in a manner similar to those described herein. All such homologous proteins and mutations thereof are included in this disclosure. The disclosure also includes combinations of naturally occurring genes and proteins, with the exception that one or more of the naturally occurring sequences may be expressed from one or more recombinant vectors. In embodiments, combinations of naturally occurring proteins, wherein the proteins are from distinct sources, are used. In embodiments, homologous proteins are from any bacteria, including but not limited to Proteobacteria. Certain embodiments of mutations in proteins that are included in the disclosure are provided below. The mutations can be in any one or any combination of proteins encoded by the *tnsA* gene, the *tnsB* gene, and the *tnsC* gene.

(47) In embodiments, the Tns proteins that are provided by this disclosure comprise mutations relative to a wild type sequence. A “wild type” sequence as used herein means a sequence that preexists in nature without experimentally engineering a change in the sequence. In embodiments, a wild type sequence is the sequence of a transposition element, a non-limiting example of which is the sequence of *Aeromonas salmonicida* strain S44 plasmid pS44-1, which can be accessed via accession no. CP022176 (Version CP022176.1), such as via

<https://www.ncbi.nlm.nih.gov/nuccore/CP022176>.

(48) In embodiments, the mutations described in i), ii) and iii) below provide for an increase in transposition frequency that is similar or greater than a value obtained from a control construct. In embodiments, the control construct comprises one or more Tns genes in which a mutation described herein is not present. In embodiments, a control transposition frequency is a frequency exhibited by a transposition element from *Aeromonas hydrophila* strain AFG_SD03, which can be identified from Accession PUTQ01000019 (Version PUTQ01000019.1), and which comprises representative amino acid sequences described below, except for the indicated mutations. The pertinent sequence of *Aeromonas hydrophila* strain AFG_SD03 can be accessed via, for example, www.ncbi.nlm.nih.gov/nuccore/1427716682. The *Aeromonas salmonicida* Cas8/5 amino acid sequence is available under accession number ASI25653, www.ncbi.nlm.nih.gov/protein/ASI25653.1; *Aeromonas salmonicida* Cas7 amino acid sequence is available under accession number ASI25654, www.ncbi.nlm.nih.gov/protein/ASI25654.1; *Aeromonas salmonicida* Cas6 amino acid sequence is available under accession number ASI25655, www.ncbi.nlm.nih.gov/protein/ASI25655.1

(49) In embodiments, for instance in bacteria, transposition frequency can be determined using, for example, a bacteriophage (i.e. viral) vector that cannot replicate or integrate into the bacterial strain used in the assay. Therefore, while the viral vector injects its DNA into the cell, it is lost during cell replication. Encoded in the phage DNA is a miniature Tn7 element where the Right and Left ends of the element flank a gene that encodes resistance to an antibiotic, such as Kanamycin (KanR). If the transposon remains on the bacteriophage DNA the cell will still be killed by the antibiotic because the bacteriophage cannot be maintained in that particular strain of bacteria. However if the TnsA, TnsB, TnsC and other required transposon proteins and nucleotide sequences described herein are added to the cell, transposition will occur because the transposon can move from the bacteriophage DNA into the chromosome (or plasmid) where it will be maintained and allow a colony of bacteria to grow that is antibiotic resistant. Therefore, when the number of infectious bacteriophage particles are in the assay is known, it permits calculation of a frequency of transposition as antibiotic resistant colonies of bacteria per bacteriophage used in the experiment. Thus, in embodiments, using one or a combination of the mutated proteins described herein increases transposition frequency, and accordingly is expected to enhance CRISPR mediated cleavage that is accompanied by the transposon-based constructs that are described herein. In alternative embodiments, detectable markers and selection elements can be used. In embodiments, transposition frequency can be measured, for example, by a change in expression in a reporter gene. Any suitable reporter gene can be used, non-limiting examples of which include adaptations of standard enzymatic reactions which produce visually detectable readouts. In embodiments, adaptations of β -galactosidase (LacZ) assays are used. In embodiments, transposition of an element from one chromosomal location to another, or from a plasmid to a chromosome, or from a chromosome to a plasmid, results in a change in expression of a reporter protein, such as LacZ. In embodiments, use of a system described herein causes a change in expression of LacZ, or any other suitable marker, in a population of cells. In embodiments, transposition efficiency is determined by measuring the number of cells within a population that experience a transposition event, as determined using any suitable approach, such as by reporter expression, and/or by any other suitable marker and/or selection criteria. In embodiments, the disclosure provides for increased transposition, such as within a population of cells, relative to a control. The control can be any suitable control, such as a reference value. In embodiments, the reference value comprises a standardized curve(s), a cutoff or threshold value, and the like. In embodiments, transposition efficiency comprises use of a system of this disclosure to transpose all or a segment of DNA from one location to another within the same or separate chromosomes, from a chromosome to a plasmid, or from a plasmid or other DNA cargo to a chromosome. In embodiments, transposition efficiency is greater than a control value obtained or derived from transposition efficiency using the

described system, but wherein the TnsA protein comprises the wild type sequence, i.e., wherein the TnsA transposon protein comprises SEQ ID NO:1, including the Alanine at position 125. In embodiments, the increased transposition efficiency using a wild type TnsA protein (i.e., a TnsA protein comprising SEQ ID NO:1) is undetectable in the same type of assay wherein a described TnsA mutant protein is used.

(50) In embodiments, assuming only for illustration, a frequency of transposition of 0.0001% is a control value because transposition efficiency was not able to be measured in the representative assays, (e.g., hypothetically only one in 100,000 cells into which a presently described system using a wild type TnsA protein experience a transposition event). In this regard, the present disclosure provides for a 1 fold to 200 fold increase in transposition efficiency, inclusive, and including all numbers and ranges of to the first decimal point there between. In embodiments, the disclosure facilitates an increase of transposition efficiency relative to a control, such as transposition from a chromosome to a plasmid, of 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, fold greater than a control value. Representative examples of assays wherein transposition frequency can be determined are described at least in FIG. 13 through FIG. 17. Similar transposition efficiency can be determined for transposition events where the transposition comprises transposing an element in cis, e.g., transposition from one location in a chromosome to a different location in the same chromosome. Representative results using this approach are shown in FIG. 17 and are described in at least Example 3. i) In one embodiment of this disclosure, the tnsA gene comprises a change in sequence such that at least one amino acid in the TnsA protein encoded by the tnsA gene is changed relative to its wild type sequence. In an embodiment, the change in the TnsA protein comprises a change of Ala at position 125 of an *Aeromonas salmonicida* TnsA protein, wherein optionally the change is to an Asp, or is a homologous change in a homologous TnsA protein. A representative TnsA amino acid sequence is provided below. In this regard, we have demonstrated that this construct can introduce numerous insertions, but without the change insertions approximate background levels, or are undetectable. ii) In embodiments, the disclosure includes a tnsB gene comprising a change in sequence such that at least one amino acid in the TnsB protein encoded by the tnsB gene is changed relative to its wild type sequence. In an embodiment, the change in the TnsB protein comprises a change of amino acid position 167 of an *Aeromonas salmonicida* TnsB protein, wherein optionally the change is a Ser, or is a homologous change in a homologous position of a homologous TnsB protein. Representative TnsB amino acid sequences are provided below. iii) As with the TnsA and TnsB proteins, in embodiments, the disclosure includes a modified tnsC gene that comprises a change in sequence such that at least one amino acid in the TnsC protein encoded by the tnsC gene is changed relative to its wild type sequence. In embodiments, the change is optionally located in a TnsC Walker B motif. In embodiments, the change in a Walker B motif is, for example, in position 135, 136, 137, 138, 139, or 140 of the *Aeromonas salmonicida* TnsC protein, a representative example of which is shown below. In one embodiment, the change is to an amino acid at position 140 in the TnsC protein, wherein, for example, amino acid 140 is change to an Ala or Gln, or a homologous change in a homologous position of a homologous TnsC protein is made. iv) the tnsC gene comprises a change in sequence such that at least one amino acid in the TnsC protein encoded by the tnsC gene is changed relative to its wild type sequence, wherein the change is optionally in a TnsC Walker B motif.

In embodiments, any composition, system, or method of this disclosure may be performed in the absence of any TnsE transposon protein. TnsE transposon proteins are known in the art. In a non-limiting embodiment, any composition, system, and/or method of this disclosure may be performed in the absence of, and/or without participation of, an *E. coli* TnsE protein that comprises or consists of the following amino acid sequence:

(51) TABLE-US-00001

MVRLATFNDNVQVVHIGHLFRNSGHKEWRIFVWFNPMQERKWTRFTHLPL
LSRAKVVNSTTKQINKADRVIEFEASDLQRAKIIDFPNLSSFASVRNKDG
AQSSFIYEAETPYKTRYHIPQLELARSLFLINSYFCRSCLSSTALQQEF
DVQYEVEVDHLEIRILPSSSFPGGALEQSAVVQLLVWLFSDQDVMDSYES
IFRHYQQNREIKNGVESWCFSFDPPPMQGWKLHVKGRRSSNEDKDYLVEEI
VGLEINAMLPSTTAISHASFQEKEAGDGSTQHIAVSTESVVDDEHLQLDD
EETANIDTDTRVIEAEPTWISFSRPSRIEKSRRARKSSQTILEKEEATTS
ENSNLVSTDEPHLGGVLAADVGGKQDATNYNSIFANRFAAFDELLSILK
TKFACRVLFEETLVLPKVGRSRLHLCKDGSPPRVIAVGVQRNGSEFVLE
VDASDGVKMLSTKVLSGVDSETWRNDFEKIRRGVVKSSLNWPNSLFDQLY
GQDGHARGVNHHPKGLGELQVSREDMEGWAERVVREQFTH.

(52) In embodiments, any composition, system, and/or method of this disclosure may be performed in the absence of, and/or without participation of any TnsE protein that is a homologue of the foregoing sequence, but is from a type of bacteria that is not *E. coli*. Non-limiting embodiments of amino acid sequences comprising mutations and/or locations of mutations are described herein, and by way of the following amino acid sequences and accession numbers. Enlarged, bold and italicized amino acids signify non-limiting examples of mutations that are encompassed by this disclosure. Enlarged sequences are locations where other mutations may be made, and are also included in this disclosure:

(53) TABLE-US-00002 TnsA (A125D) change from *Aeromonas salmonicida* strain S44 plasmid pS44-1 or TnsA (exact from *Aeromonas hydrophila* strain AFG_SD03) (SEQ ID NO: 1)

MYRRHLKHSRVKNLKFVSAKMNTVFTVESALEFDTCFHLEYSPSVKFY
EAQPEGFYEFAGRQCPYTPDFRLVDQNDVSFLEIKPSDKVADPDFLH
RFPLKQQRAIELSSPLKLVTEKQIRIcustom character PILGNLKLLHRYSGFQSFTPLH
MQLLGLVQKLGRVSLRLSDSIDAPPEEVLASALSLIARGIMQSDLTVQ
KIGISSFWAGGHSGIDHG TnsA (A125) TnsA from *Aeromonas salmonicida* strain S44 plasmidp S44-1 (SEQ ID NO: 2)
MYRRHLKHSRVKNLKFVSAKMNTVFTVESALEFDTCFHLEYSPSVKFY
EAQPEGFYEFAGRQCPYTPDFRLVDQNDVSFLEIKPSDKVADPDFLH
RFPLKQQRAIELSSPLKLVTEKQIRIcustom character PILGNLKLLHRYSGFQSFTPLHM
QLLGLVQKLGRVSLRLSDSIDAPPEEVLASALSLIARGIMQSDLTVQK
IGISSFWAGGHSGIDHG TnsB (from *Aeromonas salmonicida* strain S44 plasmidp S44-1) (SEQ ID NO: 3)

MDKHNGGLFEDEFVIPQPSTSTSPIDAIQAVLPATVDSFPYVLKVEALH
RRDYILWVEKNLAGGWTEKNLTPLLADAALVLPPTPNWRTLARWRKIY
IQHGRKLVSLIPKHQAKGNARSRLPPSDELFQEAVHRYLVGEQPSIAS
AFQLYSDSIRIENLGVVENcustom character IKTISYMAFYNRIKKLPAYQVMKSRKGSY
IADVEFKAIASHKPPSRIMERVEIDHTPLDLLLDDDLLVPLGRPSLTL
LIDAYSHCVVGFNLNFNQPSYESVRNALLSSISKDYVKNKYPSIEHEW
PCYGKPETLVVDNGVEFWASLAQSCLELGINIYNPVRKPWLKPMIER
MFGIINRKLLEPIPGKTFSNIQEKGDYDPQKDAVMRFSTFLEIFHHWVI
DVYHYEPDSRYRYIPIISWQHGNKDAPPAPIIGDDLTKLEVILSLSLHC
THRRGGIQRYHLRYDSDELASYRMNYPDQTRGKRKVLVKNLNPDISYVY

VFLEDLGSYIRVPCIDPIGYTKGLSLQEHQINVKLHRDFINEQMDVVSL
SKARIYLNDRINKNELIEVRRNIRQRNVKGVN KIAKYRNVGSHAETSIVH
ELNHPATNEVISKMESASQPEHCDDWDNFTSGLEPY TnsB (P167S) change from
Aeromonas salmonicida strain S44 plasmid pS44-1 (SEQ ID NO: 4)
MDKHNGGLFEDEFVIPQPSTSTSPIDAIQAVLPATVDSFPYVLKVEALH
RRDYILWVEKNLAGGWTEKNLTPLLADAALVLPPTPNWRTLARWRKIY
IQHGRKLVSLIPKHQAKGNARSRLPPSDELF FEQAVHRYLVGEQPSIAS
AFQLYSDSIRIENLGVVEN  custom character IKTISYMAFYNRIKKLPAYQVMKSRKGSY
IADVEFKAIAASHKPPSRIMERVEIDHTPLDLLLLDDDLLVPLGRPSLTL
LIDAYSHCVVGFNLFNFNQPSYESVRNALLSSISK KDYVKNKYP SIEHEW
PCYGKPETLVVDNGVEFWSASLAQSCLELGINIQYNPVRKPWLKPMIER
MFGIINRKLLLEPIPGKTFSNIQEKGDYDPQKDAVMRFSTFLEIFHHWVI
DVYHYEPDSRYRYIPIISWQHGNKDAPPAPIIGDDLT KLEVILSLSLHC
THRRGGIQR YHLRYDSDELASYRMNYPDQTRGKRKVLV KLNPRDISYVY
VFLEDLGSYIRVPCIDPIGYTKGLSLQEHQINVKLHRDFINEQMDVVSL
SKARIYLNDRINKNELIEVRRNIRQRNVKGVN KIAKYRNVGSHAETSIVH
ELNHPATNEVISKMESASQPEHCDDWDNFTSGLEPY TnsC (from *Aeromonas*
salmonicida strain S44 plasmid pS44-1) (SEQ ID NO: 5)
MDLSCHDADKLRSFIECYVETPLLRAIQEDFDRLRFNKQFAGEPQCMLL
TGDTGTGKSSLIRHYAAKHPEQVRHGFH HKPLLVSRI PSRPTLESTMVE
LLKDLGQFGSSDRIHKSSAESL TEALIKCLKRCETE  custom character FQELIEN
KTREKRNQIANRLKYISETAKIPIVLVGMPWATKIAEEPQWSSRLLIRR
SIPYFKLSDDRENFIRLIMGLANRMPFETQARLET KHTIYALFAACYGS
LRALKQLLDES VKQALAAHAETLKHEHIAVAYALFY PDQVNPFLQPIDE
IKACEVKQYSRYEIDAAGKEEVLNPLQFTDKIPISQLLKKR TnsC (E140A) change
from *Aeromonas salmonicida* strain S44 plasmid pS44-1 (SEQ ID NO: 6)
MDLSCHDADKLRSFIECYVETPLLRAIQEDFDRLRFNKQFAGEPQCMLL
TGDTGTGKSSLIRHYAAKHPEQVRHGFH HKPLLVSRI PSRPTLESTMVE
LLKDLGQFGSSDRIHKSSAESL TEALIKCLKRCETE  custom character FQELIEN
KTREKRNQIANRLKYISETAKIPIVLVGMPWATKIAEEPQWSSRLLIRR
SIPYFKLSDDRENFIRLIMGLANRMPFETQARLET KHTIYALFAACYGS
LRALKQLLDES VKQALAAHAETLKHEHIAVAYALFY PDQVNPFLQPIDE
IKACEVKQYSRYEIDAAGKEEVLNPLQFTDKIPISQLLKKR TnsC (E140Q) change
from *Aeromonas salmonicida* strain S44 plasmid pS44-1 (SEQ ID NO: 7)
MDLSCHDADKLRSFIECYVETPLLRAIQEDFDRLRFNKQFAGEPQCMLL
TGDTGTGKSSLIRHYAAKHPEQVRHGFH HKPLLVSRI PSRPTLESTMVE
LLKDLGQFGSSDRIHKSSAESL TEALIKCLKRCETE  custom character FQELIEN
KTREKRNQIANRLKYISETAKIPIVLVGMPWATKIAEEPQWSSRLLIRR
SIPYFKLSDDRENFIRLIMGLANRMPFETQARLET KHTIYALFAACYGS
LRALKQLLDES VKQALAAHAETLKHEHIAVAYALFY PDQVNPFLQPIDE
IKACEVKQYSRYEIDAAGKEEVLNPLQFTDKIPISQLLKKR

(54) Xre (gene 91099..91428, Locus tag CE463_00475) from *Aeromonas salmonicida* strain S44 plasmid pS44-1. The disclosure includes homologous Xre sequences. The sequence below is identical to the Xre protein in *Aeromonas hydrophila* strain AFG_SD03.

(55) TABLE-US-00003 (SEQ ID NO: 8)

MTNPLPIRLKAARKATGLTQQQLGIRLGMEQSTASARMNQYEKGKHAPDY
QTMQRIAQELGYPVAYFYCDDELLAELICMMAKLSEEKQRELLQQLSVTE YAESRDSAE

(56) In addition to any of the foregoing mutations, the disclosure also includes additional amino acid changes, such as changes in TnsC, which may include gain-of-activity mutations, in canonical Tn7 (e.g., homologous proteins), including but not necessarily limited to TnsABC(A225V),

TnsABC(E233K), TnsABC(E233A), and TnsABC(E233Q).

(57) In one aspect the disclosure includes a kit comprising one or more expression vector(s) that encodes one or more Cas or other enzymes described herein. The expression vector in certain approaches includes a cloning site, such as a poly-cloning site, such that any desirable cargo gene(s) can be cloned into the cloning site to be expressed in any target cell into which the system is introduced or already comprises. The kit can further comprise one or more containers, printed material providing instructions as to how to use make and/or use the expression vector to produce suitable vectors, and reagents for introducing the expression vector into cells. The kits may further comprise one or more bacterial strains for use in producing the components of the system. The bacterial strains may be provided in a composition wherein growth of the bacteria is restricted, such as a frozen culture with one or more cryoprotectants, such as glycerol.

(58) In another aspect the disclosure comprises delivering to cells a DNA cargo via a system of this disclosure. The method generally comprises introducing one or more polynucleotides of this disclosure, or a mixture of proteins and polynucleotides encoding the proteins, which may be also provided with RNA polynucleotides, into one or more bacterial or eukaryotic cells, whereby the Cas and transposon enzymes/proteins are expressed and editing of the chromosome or another DNA target by a combination of the Cas enzymes and the transposon occurs.

(59) In non-limiting embodiments, this disclosure is considered to be suitable for targeting eukaryotic cells, and any microorganism that is susceptible to editing by a system as described herein. In embodiments the microorganism comprises bacteria that are resistant to one or more antibiotics, whereby the editing by the present system kills or reduces the growth of the antibiotic-resistant bacteria, and/or the system sensitizes the bacteria to an antibiotic by, for example, use of cargo that targets an antibiotic resistance gene, which may be present on a chromosome or a plasmid. The disclosure is thus suitable for targeting bacterial chromosomes or episomal elements, e.g., plasmids. In embodiments, a modification of a bacterial chromosome or plasmid causes the bacteria to change from pathogenic to non-pathogenic.

(60) In embodiments, bacteria are killed. In embodiments, one or all of the components of a system described herein can be provided in a pharmaceutical formulation. Thus, in embodiments, DNA, RNA, proteins, and combinations thereof can be provided in a composition that comprises at least one pharmaceutically acceptable additive.

(61) In embodiments, the method of this disclosure is used to reduce or eradicate bacterial cells, and may be used to reduce or eradicate persister bacteria and/or dormant viable but non-culturable (VBNC) bacteria from an individual or an inanimate surface, or a food substance.

(62) In embodiments, and as noted above, the disclosure is considered suitable for editing eukaryotic cells. In embodiments, eukaryotic cells that are modified by the approaches of this disclosure are totipotent, pluripotent, multipotent, or oligopotent stem cells when the modification is made. In embodiments, the cells are neural stem cells. In embodiments, the cells are hematopoietic stem cells. In embodiments, the cells are leukocytes. In embodiments, the leukocytes are of a myeloid or lymphoid lineage. In embodiments, the cells are embryonic stem cells, or adult stem cells. In embodiments, the cells are epidermal stem cells or epithelial stem cells. In embodiments, the cells are cancer cells, or cancer stem cells. In embodiments, the cells are differentiated cells when the modification is made. In embodiments, the cells are mammalian cells. In embodiments, the cells are human, or are non-human animal cells. In embodiments, the non-human eukaryotic cells comprise fungal, plant or insect cells. In one approach the cells are engineered to express a detectable or selectable marker, or a combination thereof.

(63) In embodiments, the disclosure includes obtaining cells from an individual, modifying the cells ex vivo using a CRISPR system as described herein, and reintroducing the cells or their progeny into the individual for prophylaxis and/or therapy of a condition, disease or disorder, or to treat an injury, trauma or anatomical defect. In embodiments, the cells modified ex vivo as described herein are used autologously.

(64) In embodiments, cells modified according to this disclosure are provided as cell lines. In embodiments, the cells are engineered to produce a protein or other compound, and the cells themselves or the protein or compound they produce is used for prophylactic or therapeutic applications.

(65) In various embodiments, the modification introduced into eukaryotic cells according to this disclosure is homozygous or heterozygous. In embodiments, the modification comprises a homozygous dominant or homozygous recessive or heterozygous dominant or heterozygous recessive mutation correlated with a phenotype or condition, and is thus useful for modeling such phenotype or condition. In embodiments a modification causes a malignant cell to revert to a non-malignant phenotype.

(66) In certain aspects the disclosure includes a pharmaceutical formulation comprising one or more components of a system described herein. A pharmaceutical formulation comprises one or more pharmaceutically acceptable additives, many of which are known in the art. In some embodiments, the pharmaceutical compositions comprise a pharmaceutically acceptable carrier suitable for administration to humans. In some embodiments, the pharmaceutical compositions comprise a pharmaceutically acceptable carrier suitable for intraocular injection. In some embodiments, the pharmaceutical compositions comprise a pharmaceutically acceptable carrier suitable for topical application. In some embodiments, the pharmaceutical compositions comprise a pharmaceutically acceptable carrier suitable for intravenous injection. In some embodiments, the pharmaceutical compositions comprise and a pharmaceutically acceptable carrier suitable for injection into arteries. In some embodiments, the pharmaceutical composition is suitable for oral or topical administration. All of the described routes of administration are encompassed by the disclosure.

(67) In embodiments, a system of this disclosure is administered to an individual in a therapeutically effective amount. The therapeutically effective amount, e.g., a dose, can be estimated initially either in cell culture assays or in animal models. An animal model can also be used to determine a suitable concentration range, and route of administration. Such information can then be used to determine useful doses and routes for administration in humans, or to non-human animals. A precise dosage can be selected by in view of the patient to be treated. Dosage and administration can be adjusted to provide sufficient levels of components to achieve a desired effect, such as a modification in a threshold number of cells. Additional factors which may be taken into account include the particular gene or other genetic element involved, the type of condition, the age, weight and gender of the patient, desired duration of treatment, method of administration, time and frequency of administration, drug combination(s), reaction sensitivities, and tolerance/response to therapy. In certain embodiments, a therapeutically effective amount is an amount that reduces one or more signs or symptoms of a disease, and/or reduces the severity of the disease. A therapeutically effective amount may also inhibit or prevent the onset of a disease, or a disease relapse. In embodiments, cells modified according to this disclosure are administered to an individual in need thereof in a therapeutically effective amount.

(68) The following Examples are intended to illustrate but not limit the disclosure.

Example 1

(69) This Example demonstrates increased transposition efficiency using a mutant TnsA protein as described herein, relative to a wild type TnsA protein, and thus demonstrates that TnAs requires TnsA(A125D) for detectable RNA-guided transposition to native A.s. conjugal plasmid sequence in a heterologous host.

(70) In particular, expression of TnsABC, TniQ, Cas8/5f, Cas6f, Cas7f, and native A.s. array in *E. coli* allows RNA-guided transposition to an *Aeromonas salmonicida* (A.s.) conjugal plasmid sequence cloned onto F as monitored by a mate-out assay. Specifically, the following procedure was used.

(71) BW27783 attTnT7::miniTn7(miniTn.sup.A.s.-KanR)+F-native.sup.A.s.+pBAD33-

array.sup.A.s. strains were made chemically competent and double transformed with pBAD322-Gen TniQ/Cascade.sup.A.s. and pTA106, pTA106 TnsABC.sup.A.s. or pTA106 TnsABC.sup.A.s. (A125D) onto LB+100 µg/mL carbenicillin, 10 µg/mL gentamicin, 30 µg/mL chloramphenicol, 0.2% w/v glucose. After 24 hours incubation at 37° C., several hundred transformants were washed up in LB, diluted to a calculated OD of 0.02. 100 µL were plated on induction plates (LB+100 µg/mL carbenicillin, 10 µg/mL gentamicin, 30 µg/mL chloramphenicol, 0.2% w/v arabinose, 0.1 mM IPTG) and incubated for 16 hours at 37° C. Plates were washed up in LB, diluted to OD 6.86, and mixed 1:5 with OD 6.86 overnight culture of CW51. 20 µL drops were allowed to dry on LB plates and incubated for 4 hours at 37° C. Glass tubes were used to collect agar plugs with mated cells and 2 mLs LB were added and vortexed to extract cells from agar. Cells were serially diluted in LB+0.2% w/v glucose and plated on LB+20 µg/mL nalidixic acid, 100 µg/mL rifampicin, 100 µg/mL spectinomycin with or without 50 µg/mL kanamycin to sample the entire transconjugant population or select for transposition respectively.

(72) Transposition efficiency obtained from this approach are as follows:

(73) TABLE-US-00004 TABLE 1 Construct Transposition efficiency pTA106 (no transposon proteins, with guide 0.0000% RNA) pTA106 TnsABC-wt (full complement of 0.0000% proteins, with wild type TnsA, and guide RNA) pTA106 TnsABC-A125D (full complement 0.0188% of proteins, with TnsA A125D mutation and guide RNA)

(74) Thus, the TnsA A125D change reverts a presumably inactivating mutation in the native element.

(75) Experiments described in this Example are experimentally the same as in the following Example 2, except the TnsA A125D protein is used, along with the full complement of proteins and guide RNAs, which were targeted to lacZ-3 and lacZ-4, on opposite strands of the chromosome.

Example 2

(76) This Example, following on the design of the analysis described in Example 1, demonstrates that mutant TnAs allows RNA-guided transposition from a chromosome to distinct F plasmid targets. Specifically, expression of TnsABC (TnsA A125D), TniQ, Cas8/5f, Cas6f, Cas7f, and single repeat-spacer-repeat array units in *E. coli* allows RNA-guided transposition to distinct F bound targets as monitored by a mate-out assay. Transposition was programmed with synthetic spacers to ffs.sup.A.s. with a perfect target match (ffs.sup.exact) or with ten mismatches as found in the A.s. element (ffs.sup.wild-type) or lacZ.sup.E.c. with two randomly selected TACC PAM protospacer sequences on either strand. Results are presented in FIG. 16. Sequences used to generate the data in this figure are as follows.

(77) The segment of ffs gene from *Aeromonas salmonicida* S44 was included in the F plasmid as a target is as follows, where the protospacer is italicized and the PAM is in bold.

(78) TABLE-US-00005 (SEQ ID NO: 9)

TCAGGACCGGAAGGTAGCAGCCAAGGCGGGGGACTCGAGTGCCGGGATGT
GGCTGATGGGGCCACCAC

Guide RNA to ffs from *A. salmonicida* found in element CRISPR array (contains 10 mismatches.

(79) TABLE-US-00006 (SEQ ID NO: 10)

AGGACUGGAAGAAAUCAUCCAAGUUGGGGACU

DNA equivalent as spacer 5'-AGGACTGGAAGAAATCATCCAAGTTGGGGACT-3') (SEQ ID NO:11)

Guide RNA to ffs from *A. salmonicida* but with 10 mismatches corrected

(80) TABLE-US-00007 (SEQ ID NO: 12)

AGGACCGGAAGGUAGCAGCCAAGGCGGGGGAC

(DNA equivalent as spacer 5'-AGGACCGGAAGGTAGCAGCCAAGGCGGGGGAC-3' SEQ ID NO:13))

Sequence of the lacZ gene from *E. coli* that was included in the F plasmid as a target. Protospacer is italicized and the PAM sequences are in bold.

(81) TABLE-US-00008 (SEQ ID: 14)

CCCAATACGCAAACCGCCTCTCCCCGCGCGTTGGCCGATTCATTAATGC
AGCTGGCACGACAGGTTTCCCGACTGGAAAGCGGGCAGTGAGCGCAACG
CAATTAATGTGAGTTAGCTCACTCATTAGGCACCCCAGGCTTTACACTT
TATGCTTCCGGCTCGTATGTTGTGTGGAATTGTGAGCGGATAACAATTT
CACACAGGAAACAGCTATGACCATGATTACGGATTCACTGGCCGTCGTT
TTACAACGTCGTGACTGGGAAAACCCTGGCGTTACCCAACTTAATCGCC
TTGCAGCACATCCCCCTTTCGCCAGCTGGCGTAATAGCGAAGAGGCCCG
CACCGATCGCCCTTCCCAACAGTTGCGCAGCCTGAATGGCGAATGGCGC
TTTGCCTGGTTTCCGGCACCAAGAAGCGGTGCCGGAAAGCTGGCTGGAGT
GCGATCTTCCTGAGGCCGATACTGTCGTCGTCCCCTCAAACCTGGCAGAT
GCACGGTTACGATGCGCCCATCTACACCAACGTGACCTATCCCATTACG
GTCAATCCGCCGTTTGTTCACGAGAAATCCGACGGGTGTTACTCGC
TCACATTTAATGTTGATGAAAGCTGGCTACAGGAAGGCCAGACGCGAAT
TATTTTTGATGGCGTTAACTCGGCGTTTCATCTGTGGTGCAACGGGCGC
TGGGTGCGTTACGGCCAGGACAGTCGTTTGCCGTCTGAATTTGACCTGA
GCGCATTTTTACGCGCCGGAGAAAACCGCCTCGCGGTGATGGTGCTGCG
CTGGAGTGACGGCAGTTATCTGGAAGATCAGGATATGTGGCGGATGAGC
GGCATTTTCCGTGACGTCTCGTTGCTGCATAAACCGACTACACAAATCA
GCGATTTCCATGTTGCCACTCGCTTTAATGATGATTTTCAGCCGCGCTGT
ACTGGAGGCTGAAGTTCAGATGTGCGGCGAGTTGCGTGACTACCTACGG
GTAACAGTTTCTTTATGGCAGGGTGAAACGCAGGTGCGCCAGCGGCACCG
CGCCTTTCGGCGGTGAAATTATCGATGAGCGTGGTGTTATGCCGATCG
CGTCACACTACGTCTGAACGTCGAAAACCCGAAACTGTGGAGCGCCGAA
ATCCCGAATCTCTATCGTGCGGTGGTTGAACTGCACACCGCCGACGGCA
CGCTGATTGAAGCAGAAGCCTGCGATGTCGGTTTCCGCGAGGTGCGGAT
TGAAAATGGTCTGCTGCTGCTGAACGGCAAGCCGTTGCTGATTCGAGGC
GTTAACCGTCAAGCATCATCCTCTGCATGGTCAGGTCATGGATGAGC
AGACGATGGTGACAGGATATCCTGCTGATGAAGCAGAACAACTTTAACGC
CGTGCGCTGTTCCGATTATCCGAACCATCCGCTGTGGTACACGCTGTGC
GACCGCTACGGCCTGTATGTGGTGGATGAAGCCAATATTGAAACCCACG
GCATGGTGCCAATGAATCGTCTGACCGATGATCCGCGCTGGCTACCGGC
GATGAGCGAACGCGTAACGCGAATGGTGACGCGGATCGTAATCACCCG
AGTGTGATCATCTGGTCGCTGGGGAATGAATCAGGCCACGGCGCTAATC
ACGACGCGCTGTATCGCTGGATCAAATCTGTTCGATCCTTCCCGCCCGGT
GCAGTATGAAGGCGGCGGAGCCGACACCACGGCCACCGATATTATTTGC
CCGATGTACGCGCGCGTGGATGAAGACCAGCCCTTCCCGGCTGTGCCGA
AATGGTCCATCAAAAAATGGCTTTCGCTACCTGGAGAGACGCGCCCGCT
GATCCTTTGCGAATACGCCCACGCGATGGGTAACAGTCTTGGCGGTTTC
GCTAAATACTGGCAGGCGTTTCGTCAGTATCCCCGTTTACAGGGCGGCT
TCGTCTGGGACTGGGTGGATCAGTCGCTGATTAAATATGATGAAAACGG
CAACCCGTGGTCGGCTTACGGCGGTGATTTTGGCGATACGCCGAACGAT
CGCCAGTTCTGTATGAACGGTCTGGTCTTTGCCGACCGCACGCCGCATC
CAGCGCTGACGGAAGCAAAACACCAGCAGCAGTTTTTCCAGTTCCGTTT
ATCCGGGCAAACCATCGAAGTGACCAGCGAATACCTGTTCCGTCATAGC
GATAACGAGCTCCTGCACTGGATGGTGGCGCTGGATGGTAAGCCGCTGG
CAAGCGGTGAAGTGCCTCTGGATGTCGCTCCACAAGGTAAACAGTTGAT
TGAAGTGCCTGAACTACCGCAGCCGGAGAGCGCCGGGCAACTCTGGCTC
ACAGTACGCGTAGTGCAACCGAACGCGACCGCATGGTCAGAAGCCGGGC
ACATCAGCGCCTGGCAGCAGTGGCGTCTGGCGGAAAACCTCAGTGTGAC

GCTCCCGCCCGTCCGATCCATCTGACCCACCGACCGATG
GATTTTTGCATCGAGCTGGGTAATAAGCGTTGGCAATTTAACCGCCAGT
CAGGCTTTCTTTCACAGATGTGGATTGGCGATAAAAAACAACCTGCTGAC
GCCGCTGCGCGATCAGTTCACCCGTGCACCGCTGGATAACGACATTGGC
GTAAGTGAAGCGACCCGCATTGACCCTAACGCCTGGGTCGAACGCTGGA
AGGCGGCGGGCCATTACCAGGCCGAAGCAGCGTTGTTGCAGTGCACGGC
AGATACACTTGCTGATGCGGTGCTGATTACGACCGCTCACGCGTGGCAG
CATCAGGGGAAAACCTTATTTATCAGCCGAAAACCTACCGGATTGATG
GTAGTGGTCAAATGGCGATTACCGTTGATGTTGAAGTGGCGAGCGATAC
ACCGCATCCGGCGCGGATTGGCCTGAACTGCCAGCTGGCGCAGGTAGCA
GAGCGGGTAAACTGGCTCGGATTAGGGCCGCAAGAAAACCTATCCCGACC
GCCTTACTGCCGCTGTTTTGACCGCTGGGATCTGCCATTGTCAGACAT
GTATACCCCGTACGTCTTCCCGAGCGAAAACGGTCTGCGCTGCGGGACG
CGCGAATTGAATTATGGCCCACACCAGTGGCGCGGGCGACTTCCAGTTCA
ACATCAGCCGCTACAGTCAACAGCAACTGATGGAAACCAGCCATCGCCA
TCTGCTGCACGCGGAAGAAGGCACATGGCTGAATATCGACGGTTTCCAT
ATGGGGATTGGTGGCGACGACTCCTGGAGCCCGTCAGTATCGGCGGAAT
TCCAGCTGAGCGCCGGTCGCTACCATTACCAGTTGGTCTGGTGTCAAAA
ATAATAATAACCGGGCAGGCCATGTCTCTTTCGCTCGTTTCGCCAGCCAG
GACAGAAATGCCTCGACTTCGCTGCTGCCCAAGGTTGCCGGGTGACGCA
CACCGTGGAAACGGATGAAGGCACGAACCCAGTGGACATAAGCCTGTTC
GGTTCGTAAGCTGTAATGCAAGTAGCGTATGCGCTCACGCAACTGGTCC
AGAACCTTGACCGAACGCAGCGGTGGTAACGGCGCAGTGGCGGTTTTCA
TGGCTTGTTATGACTGTTTTTTTTGGGGTACAGTCTATGCCTCGGGCATC
CAAGCAGCAAGCGCGTTACGCCGTGGGTCGATGTTTGATGTTATGGAGC
AGCAACGATGTTACGCAGCAGGGCAGTCGCCCTAAAACAAAGTTAAACA
TCATGAGGGAAGCGGTGATCGCCGAAGTATCGACTCAACTATCAGAGGT
AGTTGGCGTCATCGAGCGCCATCTCGAACCGACGTTGCTGGCCGTACAT
TTGTACGGCTCCGCAGTGGATGGCGGCCTGAAGCCACACAGTGATATTG
ATTTGCTGGTTACGGTGACCGTAAGGCTTGATGAAACAACGCGGGCGAGC
TTTGATCAACGACCTTTTGGAAACTTCGGCTTCCCCTGGAGAGAGCGAG
ATTCTCCGCGCTGTAGAAGTCACCATTGTTGTGCACGACGACATCATTC
CGTGGCGTTATCCAGCTAAGCGCGAACTGCAATTTGGAGAATGGCAGCG
CAATGACATTCTTGCAGGTATCTTCGAGCCAGCCACGATCGACATTGAT
CTGGCTATCTTGCTGACAAAAGCAAGAGAACATAGCGTTGCCTTGGTAG
GTCCAGCGGCGGAGGAACTCTTTGATCCGGTTCCTGAACAGGATCTATT
TGAGGCGCTAAATGAAACCTTAACGCTATGGAACCTCGCCGCCCCGACTGG
GCTGGCGATGAGCGAAATGTAGTGCTTACGTTGTCCCGCATTGTTGTA
GCGCAGTAACCGGCAAAATCGCGCCGAAGGATGTCGCTGCCGACTGGGC
AATGGAGCGCCTGCCGGCCCAGTATCAGCCCGTCATACTTGAAGCTAGA
CAGGCTTATCTTGGACAAGAAGAAGATCGCTTGGCCTCGCGCGCAGATC
AGTTGGAAGAATTTGTCCACTACGTGAAAGGCGAGATCACCAAGGTAGT
CGGCAAATAATGTCTAACAATTCGTTCAAGCCGACGCCGCTTCGCGGCG
CGGCTTAACCTCAA

Guide RNA lacZ3, perfect match (SEQ ID NO: 15)
UACGGGUAACAGUUUCUUUAUGGCAGGGUGAA (DNA equivalent as spacer 5'-
TACGGGTAACAGTTTCTTTA TGGCAGGGTGAA-3' (SEQ ID NO: 16)) Guide
RNA lacZ4, perfect match (SEQ ID NO: 17)
ACAGCGGAUGGUUCGGAUAAUGCGAACAGCGC (DNA equivalent as spacer 5'-
ACAGCGGATGGTTCGGATAA TGCGAACAGCGC (SEQ ID NO: 18))

(82) To obtain the results shown in FIG. 16, the following procedures were used with the above

described spacers and guide RNAs.

(83) BW27783 attTn7::miniTn7(miniTn.sup.A.s.-KanR)+F-(target) strains were made Inoue competent and triple transformed with pTA106 TnsABC.sup.A.s.(A125D), pBAD322-Gen TniQ/Cascade.sup.A.s., and pBAD33-single array units with various guides onto LB+100 µg/mL carbenicillin, 10 µg/mL gentamicin, 30 µg/mL chloramphenicol, 0.2% w/v glucose. After 16 hours incubation at 37° C., several hundred transformants were washed up in M9 maltose. 100 µL washed up transformant pools were added to 3 mL liquid induction media (LB+100 µg/mL carbenicillin, 10 µg/mL gentamicin, 30 µg/mL chloramphenicol, 0.2% w/v arabinose, 0.1 mM IPTG) and incubated shaking for 24 hours at 30° C. 250 µL induced cultures were washed once in LB+0.2% w/v glucose and resuspended in 1 mL LB+0.2% w/v glucose. 500 µL were transferred to a glass tube and incubated shaking for 2 hours at 37° C. to allow recovery from minimal media and expression of conjugation machinery. After incubation 5 mLs mid-log CW51 (overnight culture CW51 subcultured 1:50 into LB+0.2% w/v glucose and grown for 2.5 hours) were added to each tube and incubated with gentle shaking at 37° C. to allow plasmid transfer. After 1.5 hours, cultures were vortexed to disrupt mating pairs, placed on ice, then serially diluted in LB+0.2% w/v glucose and plated on LB+20 µg/mL nalidixic acid, 100 µg/mL rifampicin, 100 µg/mL spectinomycin, 50 µg/mL X-gal, with or without 50 µg/mL kanamycin to sample the entire transconjugant population or select for transposition respectively.

Example 3

(84) This Example provides results demonstrating that Tn.sup.A.s. allows RNA-guided transposition in cis to chromosomal lacZ.

(85) In more detail, this Example uses the same transposon and Cas proteins described above, but demonstrates intra-chromosomal transposition using different targeted and control guide RNAs. In particular, the control guide RNA is directed to the ffs target used above, but the guide RNAs in this Example are programmed for guide-RNA directed editing such that they target the LacZ gene only. Results are shown in FIG. 17. In particular, the results demonstrate that expression of TnsABC (TnsA A125D), TniQ, Cas8/5f, Cas6f, Cas7f, and single repeat-spacer-repeat array units in *E. coli* allows RNA-guided transposition to lacZ as monitored by Lac phenotype on MacConkey agar. Transposition can thus be programmed with synthetic spacers to lacZ with two randomly selected TACC PAM protospacer sequences on either strand. To obtain the results described in this Example, the following materials and methods were used.

(86) BW27783 Lac+attTn7::miniTn7(miniTnAs-KanR) was made calcium competent and triple transformed with pTA106 TnsABC.sup.As(A125D), pBAD322-Gen TniQ/Cascade.sup.As, and pBAD33-single array units with various guides onto LB+100 µg/mL carbenicillin, 10 µg/ml gentamicin 30 µg/mL chloramphenicol, 0.2% w/v glucose. After 16 hours incubation at 37° C., several hundred transformants were washed up in M9 maltose. Cells were diluted in 3 mL liquid induction media (LB+100 µg/mL carbenicillin, 10 µg/mL gentamicin, 30 µg/L chloramphenicol, 0.2% w/v arabinose, 0.1 mM IPTG) to a calculated OD 0.2. After 24 hour incubation at 30° C., cells were diluted and plated on Mac Lac. After 16 hours incubation at 37° C., white and red colonies were counted. Insertions directed into the lacZ are indicated with a color change on indicator media, wherein a Lac-colony is white and a Lac+ is red on MacConkey's media. The pBAD33-single array units encoding lacZ3, lacZ4, and ffs.sup.wild-type guide RNAs were the same as used in Example 2. The *E. coli* native lacZ gene targeted in these is shown in Example 2. There was no match to the ffs.sup.wild-type guide RNAs so that it could act as a negative control. FIG. 18 depicts pertinent segments of the guide RNAs for each gene, and the target sequences in relation to the TSD and the right end.

(87) While the disclosure has been particularly shown and described with reference to specific embodiments, it should be understood by those having skill in the art that various changes in form and detail may be made therein without departing from the spirit and scope of the present disclosure as disclosed herein.

Claims

1. A method for modifying a chromosome or extrachromosomal element in one or more prokaryotic cells, the method comprising introducing into the one or more prokaryotic cells a TnsA transposon protein that is at least 90% similar to SEQ ID NO:1 but contains an amino acid other than Alanine at position 125 of SEQ ID NO:1; or an expression vector comprising a *tnsA* gene that encodes and expresses said TnsA transposon protein when introduced into the one or more prokaryotic cells; wherein the one or more prokaryotic cells also comprise during said modifying: transposon proteins TnsB, TnsC, TniQ, Cas proteins Cas8f, Cas5f, Cas7f, Cas6f, a transposable DNA cargo sequence that is flanked by left and right transposon sequences; and at least one guide RNA comprising a spacer targeted to a target DNA sequence in the chromosome or the extrachromosomal element, the DNA target sequence comprising a protospacer and a protospacer adjacent motif (PAM) that is 5' to the protospacer; wherein the TnsA, TnsB, TnsC, TniQ, Cas8f, Cas5f, Cas7f, Cas6f, and the at least one guide RNA, participate in transposition of the transposable DNA cargo sequence into the chromosome or extrachromosomal element in a location that is 3' relative to the PAM and the protospacer.
2. The method of claim 1, wherein the TnsA transposon protein is at least 90% similar to SEQ ID NO:1 and comprises a D at position 125 of SEQ ID NO:1.
3. The method of claim 2, wherein the TnsA transposon protein comprises SEQ ID NO:2.
4. The method of claim 1, wherein efficiency of transposition of the transposable DNA cargo in a population of prokaryotic cells is more efficient than transposition obtained from a control using a TnsA transposon protein that comprises an amino acid sequence that is at least 90% similar to SEQ ID NO:1 and contains an Alanine at position 125.
5. The method of claim 4, wherein at least one of the TnsA, TnsB, TnsC, TniQ, Cas8f, Cas5f, Cas7f, and Cas6f proteins is expressed in the one or more prokaryotic cells from an expression vector.
6. The method of claim 5, wherein the TnsA protein is expressed from an expression vector, and wherein at least one of TnsB, TnsC, TniQ, Cas8f, Cas5f, Cas7f, Cas6f, is also expressed from the same expression vector.
7. The method of claim 6, wherein the TnsA, TnsB, TnsC, TniQ, Cas8f, Cas5f, Cas7f, Cas6f, are all expressed from the same expression vector.
8. The method of claim 5, wherein the at least one guide RNA is introduced into the one or more prokaryotic cells as an RNA polynucleotide.
9. The method of claim 5, wherein the at least one guide RNA is expressed from an expression vector within the one or more prokaryotic cells.
10. The method of claim 9, wherein the expression vector that expresses the guide RNA also comprises the transposable DNA cargo.
11. The method of claim 9, wherein the expression vector that expresses the at least one guide RNA is distinct from an expression vector that expresses at least one of the TnsA, TnsB, TnsC, TniQ, Cas8f, Cas5f, Cas7f, and Cas6 proteins.
12. The method of claim 10, wherein the expression vector that expresses the at least one guide RNA is distinct from an expression vector that expresses at least one of the TnsA, TnsB, TnsC, TniQ, Cas8f, Cas5f, Cas7f, and Cas6 proteins.
13. The method of claim 1, wherein the TnsA, TnsB, TnsC, TniQ, cas8f, cas5f, cas7f, and cas6f proteins are expressed within the one or more prokaryotic cells from the same expression vector, wherein the at least one guide RNA is expressed from an expression vector within the one or more prokaryotic cells that is distinct from the expression vector that expresses the TnsA, TnsB, TnsC, TniQ, cas8f, cas5f, cas7f and cas6f proteins, wherein the expression vector that expresses the guide RNA also comprises the transposable DNA cargo, and wherein TnsA protein comprises the

sequence of SEQ ID NO:2.

14. The method of claim 13, wherein the transposable DNA cargo sequence is transposed into the chromosome or extrachromosomal element wherein the transposable DNA cargo sequence is transposed into the chromosome or extrachromosomal element within 42-52 nucleotides 3' relative to the 3' end of the protospacer.
